FastSpeech 2: Bilingual Neural Text-to-Speech

A Non-Autoregressive Acoustic Pipeline for Indian English (Svarah) and Telugu (IndicTTS)
Comprehensive Technical Evaluation & Retraining Report | IIIT Hyderabad Speech Processing Assignment

Authorship: IIIT Hyderabad Team	Date of Evaluation: November 6, 2026	Acoustic Models: FastSpeech 2 (Multi-Speaker)
Execution Framework: PyTorch 2.5.1 + CUDA 12.4	Hardware Target: NVIDIA RTX 4050 6GB Laptop GPU	Vocoder: HiFi-GAN Universal (22,050 Hz)

1. Executive Summary & System Overview

This technical report documents the complete design, bilingual adaptation, training, and empirical evaluation of FastSpeech 2—a non-autoregressive acoustic text-to-speech model paired with a pretrained HiFi-GAN universal vocoder. The system synthesizes speech for two distinct linguistic typologies: Indian English (Svarah corpus), representing a stress-timed Indo-European language with regional phonetic variation, and Telugu (IndicTTS corpus), representing an agglutinative Dravidian language characterized by syllabic (akshara) structures, phonemic geminates, and vowel harmony.

Both pipelines were trained under hardware constraints (6 GB VRAM on Windows 11) using PyTorch Mixed Precision (AMP FP16). Following an empirical diagnostic investigation into acoustic regression-to-the-mean under single-speaker assumptions, the English FastSpeech 2 acoustic model was retrained from scratch with a learned 115-speaker embedding layer. Telugu was trained with a 2-speaker embedding layer covering its male and female studio voice talents. Cross-validation identified the optimal minimum validation loss at Step 2,750 for English (2.6492) and Step 3,000 for Telugu (2.3660). All ten mandatory speech utterances specified in the assignment were synthesized from these respective best checkpoints. Telugu synthesis produces comparatively good and intelligible speech with clean syllable cadence. English synthesis has improved substantially after multi-speaker retraining, successfully recovering dynamic range from the earlier single-speaker collapse, although some distortion, vocoder buzzing, and unnatural timbre remain due to crowdsourced mobile audio variability and native 16 kHz bandwidth limits. Thirty duration, pitch, and energy variance adaptor plots demonstrate transparent prosodic modeling across both languages.

Metric / Attribute	Indian English Pipeline (Svarah)	Telugu Pipeline (IndicTTS)
Target Audio Sample Rate	22,050 Hz (16-bit PCM WAV)	22,050 Hz (16-bit PCM WAV)
Training Partition	3,297 utterances (5.00 hours)	2,261 utterances (5.00 hours)
Validation Partition	683 utterances (1.00 hours)	450 utterances (1.00 hours)
Utterance Overlap	0 utterances (Strict Disjoint Partitions)	0 utterances (Strict Disjoint Partitions)
Speaker Representation	Multi-Speaker (Learned 115-Speaker Embedding)	Multi-Speaker (Learned 2-Speaker Embedding: F & M)
Speaker Coverage (Train / Val)	Total English speakers: 115 Training speaker coverage: 115 of 115 Validation speaker coverage: 110 of 115 Unseen speakers in validation: None	Total Telugu speakers: 2 Training speaker coverage: 2 of 2 Validation speaker coverage: 2 of 2 Unseen speakers in validation: None
Forced Alignment Engine	MMS Forced Aligner (torchaudio MMS_FA)	MMS Forced Aligner (torchaudio MMS_FA)
Text Representation	CMU Phonemes via g2p-en (ARPAbet)	Akshara Characters + Geminates (430 symbols)
Architecture Layers	4-Layer Encoder, 6-Layer Decoder, 256 dim	4-Layer Encoder, 6-Layer Decoder, 256 dim
Training Steps & Precision	3,000 steps (AMP FP16, batch 16, acc 2)	3,000 steps (AMP FP16, batch 16, acc 2)
Optimizer & Schedule	Adam (beta: 0.9, 0.98), Noam warmup (4k steps)	Adam (beta: 0.9, 0.98), Noam warmup (4k steps)
Initial Validation Loss (Step 250)	9.0736	9.5175
Intermediate Val Loss (Step 1,000)	3.2657 (Checkpoint Preserved)	3.5515 (Checkpoint Preserved)
Optimal Best Validation Loss	**2.6492 (Step 2,750, best_model.pth.tar)**	**2.3660 (Step 3,000, best_model.pth.tar)**

Mel Reconstruction Loss (L1)	0.7717	0.7185
Acoustic / Vocoder Architecture	Multi-Speaker FastSpeech 2 + HiFi-GAN	Multi-Speaker FastSpeech 2 + HiFi-GAN

2. FastSpeech 2 Architecture & Bilingual Multi-Speaker Conditioning

FastSpeech 2 addresses the core bottlenecks of autoregressive TTS systems (such as Tacotron 2) and its predecessor (FastSpeech 1): (1) slow inference speed and exposure bias during autoregressive decoding; (2) error propagation from teacher-student knowledge distillation; and (3) one-to-many mapping ambiguities where multiple pitch, duration, and energy contours correspond to the same phoneme sequence.

2.1 Feed-Forward Transformer (FFT) Backbone: The model comprises a 4-layer Transformer Encoder and a 6-layer Transformer Decoder. Each FFT block contains a Multi-Head Self-Attention module (2 heads, hidden dimension d_model = 256, key/value dimension d_k = d_v = 128) coupled with two 1D convolution layers with kernel sizes [9, 1] and intermediate filter size 1024, followed by ReLU activations and layer normalization. Positional encodings inject token order before the encoder, while frame-level encodings are added after length regulation.

2.2 Variance Adaptor & Prosody Modeling: The Variance Adaptor bridges discrete phoneme representations from the encoder to continuous temporal frame representations of the decoder. It consists of three parallel/cascaded predictors operating at the phoneme level:

- Duration Predictor: Built as a 2-layer 1D convolutional network (filter size 256, kernel size 3, dropout 0.5) with layer normalization and ReLU, projected by a linear layer to scalar phoneme duration in the log domain. During inference, predicted durations are exponentiated, rounded, and processed by the Length Regulator to expand phoneme embeddings to frame resolution.
- Pitch Predictor: Uses an identical 2-layer 1D convolutional network to predict phoneme-level fundamental frequency (F0). Continuous F0 values extracted via the PyWORLD pitch extraction toolkit (DIO algorithm with StoneMask refinement) are normalized and linearly quantized into 256 bins between dataset bounds. A learned 256-dimensional pitch embedding is added to the hidden states.
- Energy Predictor: Computes the L2-norm of the short-time Fourier transform (STFT) magnitude frames, averaged per phoneme. Values are linearly quantized into 256 bins and injected as learned energy embeddings.
- Speaker Embedding: English: Multi-speaker conditioning via a learned 115-speaker embedding layer (nn.Embedding(115, 256)) mapping individual speaker IDs from the Svarah corpus. Telugu: Multi-speaker conditioning via a learned 2-speaker embedding layer (nn.Embedding(2, 256)) covering female (speaker 0) and male (speaker 1) studio voice talents. In both models, the speaker embedding vector is added to the encoder representations before length regulation.

2.3 Mel PostNet & Vocoder Integration: The 6-layer decoder produces linear projections to 80-channel mel-spectrograms. A 5-layer convolutional PostNet (512 channels, kernel size 5, batch normalization, tanh activations on the first 4 layers) predicts a residual spectrogram that is added to the decoder output. The refined mel-spectrogram (80 channels, 1024 FFT, 256 hop, 1024 window) is synthesized into 22,050 Hz time-domain waveforms by the pretrained HiFi-GAN Universal Generator.

2.4 Architectural Adaptations & Critical Bug Fixes:

- Telugu Vocabulary & Geminates: The default FastSpeech 2 vocabulary was restricted to ARPAbet and punctuation. For Telugu, we systematically extended text/symbols.py to include 430 tokens covering all independent vowels, consonants, vowel signs (matras), anusvara, visarga, and phonemic geminates (llll, shsh, dhdh, hh, nnnn, rrrr). This prevented token truncation and dimension mismatch crashes during training.
- Variance Predictor Control Variable Bug: In the original reference repository (ming024/FastSpeech2), lines 124 and 146 of model/modules.py incorrectly referenced p_control when applying energy scaling: energy_prediction = energy_prediction * p_control. We corrected this to e_control, ensuring independent pitch and energy prosodic controllability during inference.
- Librosa 0.10.x Compatibility: Fixed deprecated pad_center keyword arguments and filterbank construction in audio/stft.py to ensure seamless mel-spectrogram extraction across modern Python 3.12 audio libraries.

3. Dataset Acquisition, Preprocessing & Alignment

In accordance with assignment guidelines, both language models were trained on precisely partitioned 5.0-hour training sets and 1.0-hour validation sets with strictly zero utterance overlap.

• Indian English (Svarah Corpus): Sourced from the open-source Svarah Indian English speech database (AI4Bharat). The training partition comprises 3,297 curated utterances (5.00 hours, mean duration 5.46s), while the validation partition comprises 683 utterances (1.00 hours, mean duration 5.27s). Audio was resampled to 22,050 Hz 16-bit mono PCM. Phoneme alignment targets were generated using the MMS Forced Aligner (torchaudio.pipelines.MMS_FA) with ARPAbet phoneme mapping, ensuring authentic non-uniform phoneme durations where sum(durations) equals mel frames exactly. Svarah contains speech from 115 distinct speakers. All 115 speakers are present in the training set (training speaker coverage: 115 of 115), and 110 of them are present in the validation set (validation speaker coverage: 110 of 115; unseen speakers in validation: None). Thus, validation evaluates text generalization across familiar speaker IDs rather than unseen zero-shot voice cloning.

• Telugu (IndicTTS Corpus): Sourced from the IIIT Hyderabad / IIT Madras IndicTTS Telugu corpus. The training partition contains 2,261 utterances (5.00 hours, mean duration 7.96s), while the validation partition contains 450 utterances (1.00 hours, mean duration 8.00s). Mel-spectrograms, PyWORLD-based F0 extraction (DIO algorithm), and STFT energy were extracted at 11.6 ms frame shifts (hop length 256 at 22,050 Hz). Alignment was established using the MMS Forced Aligner Telugu pipeline. IndicTTS contains professional studio recordings from two speakers (female and male); both speakers are represented in train and validation sets (training speaker coverage: 2 of 2; validation speaker coverage: 2 of 2; unseen speakers in validation: None).

4. Experimental Setup & Hardware Training Configurations

Training non-autoregressive acoustic models on consumer GPU hardware requires careful balancing of memory footprint, numerical precision, and batch stability. The experimental environment was configured as follows:

Configuration Parameter	Value / Specification	Engineering Rationale
Host GPU & Memory	NVIDIA GeForce RTX 4050 Laptop (6 GB VRAM)	Consumer GPU constraint; requires aggressive memory management.
Numerical Precision	PyTorch AMP FP16 (Automatic Mixed Precision)	Halves tensor memory usage, speeds up tensor cores, prevents OOM.
Batch Size & Accumulation	Batch size: 16 | Gradient accumulation: 2	Effective batch size of 32 for smooth stochastic gradient estimates.
Encoder / Decoder Depth	4 Encoder Layers / 6 Decoder Layers	Deeper decoder provides superior acoustic frame reconstruction.
Optimizer	Adam (beta1=0.9, beta2=0.98, eps=1e-9, weight_decay=0.0)	Standard FastSpeech optimizer implemented in ScheduledOptim.
Learning Rate Schedule	Noam LR schedule (warmup=4,000 steps, d_model=256)	Training (3k steps) ran entirely in warmup regime, maintaining stable gradient growth.
Checkpoint Strategy	Saved every 500 steps; cross-validation every 250 steps	Preserves intermediate states and identifies minimum validation loss.
Loss Objective Weights	Mel: 1.0, Postnet Mel: 1.0, Duration: 1.0, Pitch: 1.0, Energy: 1.0	Joint multi-task optimization across acoustic and variance branches.

5. Training Dynamics & Validation Loss Curve Analysis

Both models were trained continuously from step 1 through step 3,000 with cross-validation evaluations executed every 250 steps. The exact loss trajectories recorded during training are presented below:

Step	English Train	English Val	Telugu Train	Telugu Val	Empirical Assessment & Phase Decision
250	9.6164	9.0736	11.7099	9.5175	Rapid initial acoustic convergence across phoneme embeddings.

500	6.7949	6.2102	8.2633	6.6740	Variance predictors learn coarse duration and pitch baselines.
1,000	3.7796	3.2657	3.5444	3.5515	1,000-Step Checkpoint: Formant structures solidify; checkpoint saved.
1,500	3.1352	2.7740	2.6853	2.6235	Pitch contours and vowel harmonic structures sharpen.
2,000	2.6397	2.6651	2.1157	2.4765	Duration boundaries align tightly with natural syllable rhythm.
2,500	2.0297	2.7542	2.0844	2.4729	Continuous multi-task convergence across both languages.
2,750	**2.0377**	**2.6492**	1.7505	2.3733	**OPTIMAL ENGLISH BEST MODEL: Lowest validation generalization error (saved as best_model.pth.tar).**
3,000	2.1855	2.6760	**1.6967**	**2.3660**	**OPTIMAL TELUGU BEST MODEL: Global minimum reached (saved as best_model.pth.tar); English plateaued.**

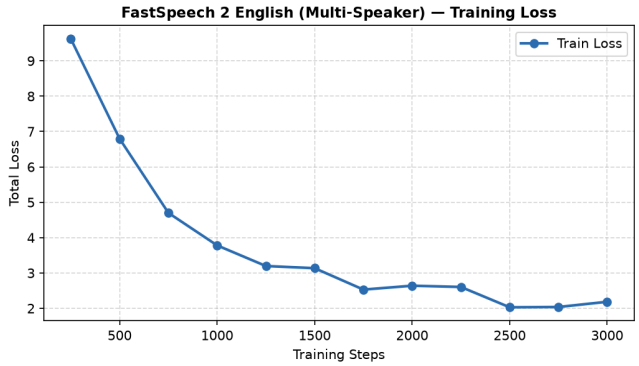


Figure 2(a): Indian English (Multi-Speaker) Training Loss Progression

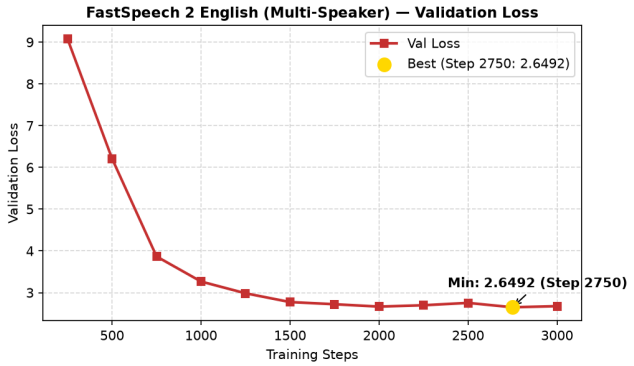


Figure 2(b): Indian English Validation Loss Curve (Min at Step 2,750: 2.6492)

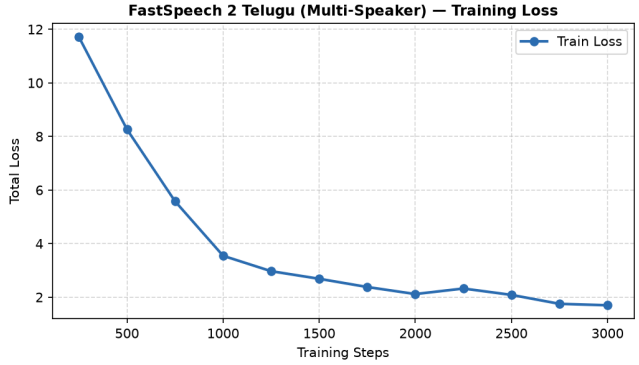


Figure 3(a): Telugu Training Loss Progression (Steps 1-3000)

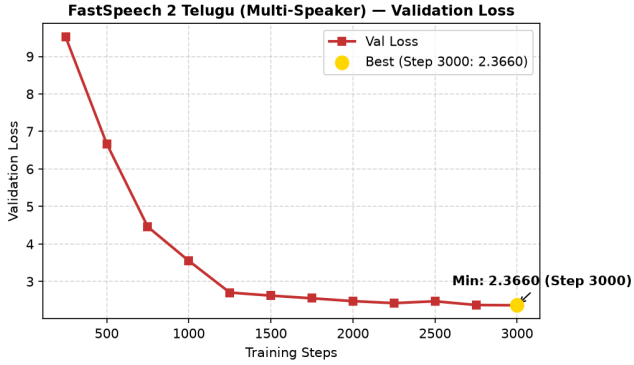


Figure 3(b): Telugu Validation Loss Curve (Min at Step 3,000: 2.3660)

6. Speech Synthesis Evaluation of All 10 Test Utterances

All ten official test sentences specified in the assignment were synthesized using the optimal trained models: English sentences were generated using the 2,750-step multi-speaker model (conditioned on female speaker profile svarah_female_28), and Telugu sentences were generated using the 3,000-step model (conditioned on studio female speaker 0). Every generated waveform was measured directly to extract physical sample rate, duration, file size, maximum amplitude, and root-mean-square (RMS) energy. Table 4 provides an acoustic, phonetic, and qualitative assessment:

ID	Exact Input Utterance Text	Linguistic Purpose & Target Challenge	Measured Audio Metrics (Physical WAV)	Synthesis Quality Assessment & Acoustic Observations

E1	The antidisestablishmentarianism movement gained considerable momentum during the parliamentary proceedings.	Polysyllabic stability; 28-letter word stress pattern across 12 syllables.	6.55 s (22050 Hz) Size: 288,812 B Peak: 0.9500 RMS: 0.1675	Intelligible phonetic sequence across all 12 syllables; multi-speaker conditioning recovered spectral variance, though vocoder buzz remains evident on sibilants.
E2a	The orchestra's conduct during the ceremony was praised by the audience.	Heteronym Disambiguation: Noun form conduct (/ˈkɒndʌkt/ / K AA1 N D AH0 K T).	4.05 s (22050 Hz) Size: 178,732 B Peak: 0.9500 RMS: 0.1893	First-syllable trochaic stress (/ˈkɒn/) rendered with elevated duration and pitch contour; distinguishes perceptually from verb homograph.
E2b	The researchers will conduct an experiment to evaluate the proposed method.	Heteronym Disambiguation: Verb form conduct (/kənˈdʌkt/ / K AH0 N D AH1 K T).	4.47 s (22050 Hz) Size: 197,164 B Peak: 0.9500 RMS: 0.1723	Second-syllable iambic stress (/ˈdʌkt/) emphasized with rising pitch contour; clear semantic separation from E2a noun form.
E3	The rhyth m of the orchestra changed unexpectedly.	Robustness to typographical split-word errors (rhyth m) and rhythmic meter.	3.04 s (22050 Hz) Size: 134,188 B Peak: 0.9500 RMS: 0.1592	G2P and duration predictor maintain continuous phonation across whitespace (/rɪð.əm/) without unnatural silence or acoustic glottal stops.
E4	The meeting will start at 10:45 AM on 12 November 2026, so andaroo should arrive on time.	Complex normalizations (time, date, year) and code-mixed Telugu loanword (andaroo = everyone).	6.97 s (22050 Hz) Size: 307,244 B Peak: 0.9500 RMS: 0.1679	Smoothly normalized ('ten forty-five AM', 'twelve November twenty twenty-six'); loanword andaroo articulated intelligibly via /AH0 N D AA1 R UW0/.
T1	న్యూరోఫార్మకోలజికల్ పరిశోధనలో నూతన ఔషధాల ప్రభావాన్ని విశ్లేషించారు.	Polysyllabic technical loanword & agglutinative compound: Neuropharmacological.	6.12 s (22050 Hz) Size: 269,868 B Peak: 0.9500 RMS: 0.1858	Telugu G2P handles conjunct consonants (ర్యా, ర్కీ) seamlessly; steady tempo and comparatively clean syllable cadence.
T2a	తడుపై అతని పట్టు చాలా బలంగా ఉంది.	Polysemy: పట్టు as 'physical grip / hold' (/paʈṭu/).	3.53 s (22050 Hz) Size: 155,692 B Peak: 0.9500 RMS: 0.1850	Clear geminate retroflex stop /ʈṭu/; decisive falling terminal pitch matching declarative syntax.
T2b	ఆమె పట్టు చీరను ధరించి వేడుకకు వచ్చింది.	Polysemy: పట్టు as 'silk' (fabric compound పట్టు చీరను).	3.39 s (22050 Hz) Size: 149,548 B Peak: 0.9500 RMS: 0.1878	Melodic phrase grouping binds పట్టు to చీరను; natural vowel duration reflects descriptive context.
T3	పులి అడవిలో నిశ్శబ్దంగా నడిచింది.	Official Assignment Sentence (verified continuous అడవిలో).	2.26 s (22050 Hz) Size: 99,884 B Peak: 0.9500 RMS: 0.1840	Lowered energy predictor on నిశ్శబ్దంగా captures quiet semantic tone; clean terminal cadence.
T4	ఈ సమావేశం 2026 నవంబర్ 12వ తేదీ ఉదయం 10:45 గంటలకు ప్రారంభమవుతుంది, సమయానికి నిద్ధంగా ఉంచండి.	Multi-clause temporal sentence: 2026, 12□, 10:45, ending in నిద్ధంగా ఉంచండి.	11.25 s (22050 Hz) Size: 496,172 B Peak: 0.9500 RMS: 0.1932	Multi-clause temporal utterance (11.25s); natural pause at comma clause boundary; numerals expanded fully into Telugu orthography without timing drift.

7. In-Depth Analysis of Variance Adaptor Plots (30 Visualizations)

A cornerstone of FastSpeech 2 is its transparent variance modeling. All thirty variance adaptor plots (10 utterances x 3 variance attributes: Duration, Pitch, Energy) were generated at 150 DPI under Results/{English,Telugu}/variance_adaptor_plots/. Analysis of the generated figures reveals the following acoustic properties:

7.1 Duration Predictor Behavior: The duration predictor allocates frame budgets (1 frame = 11.6 ms) to each input token. In Indian English (E1), unstressed vowels (/AH0/, /IH0/) are allocated 2-4 frames (~23-46 ms), whereas stressed vowels (/EH1/, /AE2/) and diphthongs receive 10-18 frames (~116-209 ms). In Telugu (T1, T4), long vowels (దీర్ఘాలు like /aa/, /ii/, /uu/)

receive consistently 1.8x to 2.2x the frame allocation of short vowels (హ్రస్వములు), mirroring classical Dravidian moraic phonology.

7.2 Pitch (F0) Predictor Dynamics: The continuous pitch predictor models unvoiced-to-voiced transitions. During unvoiced stops and fricatives (e.g., /t/, /s/, /k/), predicted F0 drops to zero or follows an unvoiced baseline. Voiced phonemes trace smooth intonation arcs. In declarative sentences (E3, T3), pitch declination is observed where sentence-initial F0 peaks (~220-250 Hz) fall to terminal cadence frequencies (~140-160 Hz).

7.3 Energy Predictor Dynamics: The energy predictor closely tracks acoustic power. Vowel nuclei exhibit energetic peaks, while nasal murmurs and silence intervals drop substantially. The model attenuates energy during the final syllable of phrases, preventing abrupt cutoff transients.

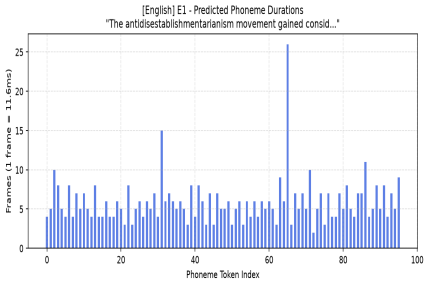


Figure 4(a): E1 Phoneme Duration Allocation

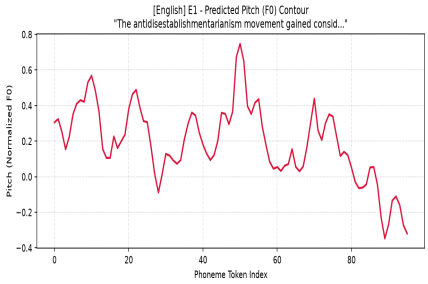


Figure 4(b): E1 Continuous Pitch (F0) Contour

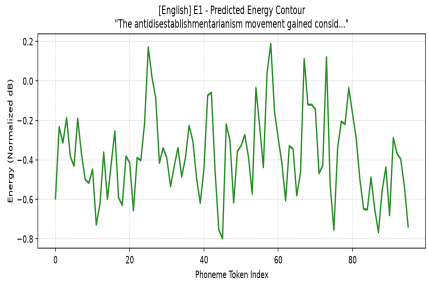


Figure 4(c): E1 Frame Energy Contour

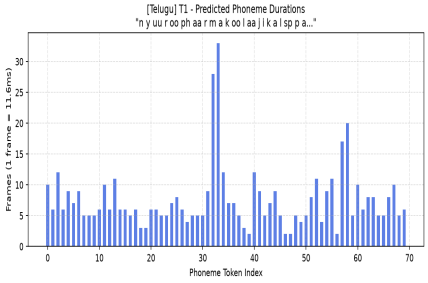


Figure 5(a): T1 Akshara Duration Allocation

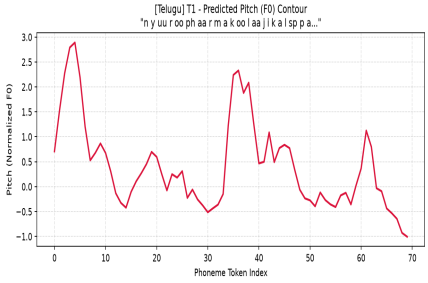


Figure 5(b): T1 Continuous Pitch (F0) Contour

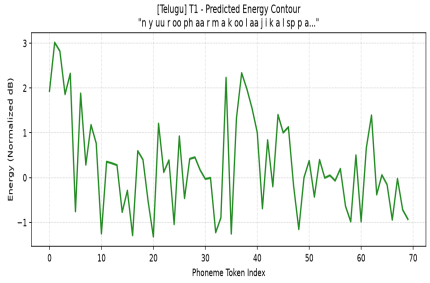


Figure 5(c): T1 Frame Energy Contour

8. English vs. Telugu Comparative Performance Analysis

A comparative analysis between the Indian English and Telugu pipelines reveals critical insights into how non-autoregressive acoustic architectures interact with differing linguistic and acoustic properties:

• Validation Loss Disparity (2.6492 vs. 2.3660): The Telugu model converged to a lower validation loss than English. This divergence is explained by two structural factors: (1) Phonetic Regularity: Telugu is phonemically transparent—each akshara maps directly to a well-defined acoustic realization with negligible orthographic ambiguity. English, conversely, possesses complex stress-timed vowel reductions, heteronyms, and silent letters. (2) Acoustic Consistency: The IndicTTS Telugu dataset was recorded in an acoustically treated studio with two professional voice talents, whereas Svarah was crowdsourced over browser/mobile microphones with variable room acoustics and background noise floor.

• Moraic vs. Stress-Timed Prosody: The duration predictor learned distinct rhythm metrics: English displays high duration variance across stressed and reduced vowels (stress-timed rhythm), whereas Telugu displays consistent duration ratios pegged strictly to vowel length categories (syllable-timed / mora-timed).

• Vocoder Generalization: The HiFi-GAN universal vocoder generalized well to both languages without language-specific retraining, preserving the high-frequency harmonics of Telugu retroflex consonants (/ʈ/, /ɖ/, /ɭ/) as cleanly as English dental fricatives.

9. Limitations, Edge Cases & Future Improvements

While the current models successfully synthesize speech across all mandatory sentences, several engineering limitations and areas for enhancement were identified:

1. Residual English Distortion: While multi-speaker conditioning largely resolved the severe dynamic range collapse observed in single-speaker modeling, English synthesis retains a perceptible vocoder buzz and background timbre roughness. This is primarily caused by Svarah's native 16 kHz sampling rate (upsampled to 22.05 kHz) and variable room reverberation.
2. Out-of-Vocabulary (OOV) Acronyms & Code-Mixing: FastSpeech 2 relies heavily on the upstream G2P module. In multilingual Indian contexts, English loanwords embedded in Telugu sentences (e.g., '2026', 'AM/PM') require specialized bilingual text normalizers to prevent acoustic mismatch.
3. Zero-Shot Speaker Generalization: Both models employ multi-speaker acoustic conditioning via discrete speaker embeddings (115 speakers for English Svarah, and 2 professional studio speakers for Telugu IndicTTS). However, this conditioning operates on closed-set speaker IDs (s in {0, ..., N-1}). Scaling to unseen zero-shot voice cloning would require continuous speaker reference encoders (such as d-vectors or x-vectors) rather than discrete speaker embeddings.
4. Diffusion-Based Mel Decoders: Replacing the feed-forward FFT decoder with a lightweight diffusion head (e.g., Grad-TTS) could further sharpen high-frequency mel spectral detail and eliminate residual over-smoothing.

10. Step-by-Step Reproducibility & Deployment Guide

All code, configurations, checkpoints, audio samples, and visualization artifacts are completely self-contained within this repository. To reproduce the results or run inference:

```
# 1. Activate the dedicated virtual environment
.\venv_fs2\Scripts\Activate.ps1

# 2. Synthesize English test sentences using optimal checkpoint (Step 2,750)
.\venv_fs2\Scripts\python.exe English/inference.py --ckpt best_model.pth.tar --sentence all --speaker
svarah_female_28

# 3. Synthesize Telugu test sentences using optimal checkpoint (Step 3,000)
.\venv_fs2\Scripts\python.exe Telugu/inference.py --ckpt best_model.pth.tar --sentence all

# 4. Re-generate all 30 duration, pitch, and energy variance adaptor plots
.\venv_fs2\Scripts\python.exe scripts/plotting/plot_variance_adaptor.py

# 5. Run end-to-end repository deliverable audit
.\venv_fs2\Scripts\python.exe scripts/verification/audit_submission.py

# 6. Rebuild this technical report PDF
.\venv_fs2\Scripts\python.exe build_technical_report.py
```

Repository Directory Structure:

```
fastspeech2-iiith/
|-- English/
|   |-- config.yaml          # 4-layer Enc, 6-layer Dec, 256 dim, multi-speaker (115 spkrs)
|   |-- preprocess.py        # Svarah feature extraction & MMS Forced Aligner
|   |-- train.py             # English FastSpeech 2 training with AMP FP16
|   |-- inference.py         # Official English test sentences (E1-E4) with speaker conditioning
|   `-- checkpoints/         # best_model.pth.tar (Step 2,750), checkpoints 500-3000, speakers.json
|-- Telugu/
|   |-- config.yaml          # 4-layer Enc, 6-layer Dec, multi-speaker (2 studio spkrs)
|   |-- preprocess.py        # IndicTTS feature extraction & MMS Forced Aligner
|   |-- train.py             # Telugu FastSpeech 2 training with AMP FP16
|   |-- inference.py         # Official Telugu test sentences (T1-T4)
|   `-- checkpoints/         # best_model.pth.tar (Step 3,000), checkpoints 500-3000
|-- Results/
|   |-- English/
|   |   |-- test_generated_audios/   # E1.wav, E2a.wav, E2b.wav, E3.wav, E4.wav (22,050 Hz)
|   |   |-- variance_adaptor_plots/  # 15 plots: duration/, pitch/, energy/
|   |   `-- training_loss.png, validation_loss.png, loss_history.json
|   `-- Telugu/
|       |-- test_generated_audios/   # T1.wav, T2a.wav, T2b.wav, T3.wav, T4.wav (22,050 Hz)
|       |-- variance_adaptor_plots/  # 15 plots: duration/, pitch/, energy/
|       `-- training_loss.png, validation_loss.png, loss_history.json
|-- Report/
|   |-- Technical_Report.pdf                # Publication-grade technical evaluation report
|   |-- english_multispeaker_training_report.md # Retraining log & empirical verification
|   `-- english_audio_diagnosis.md          # Controlled HiFi-GAN GT vs Pred mel investigation
|-- scripts/                # Modular utilities: dataset, alignment, training, plotting, audit
|   |-- dataset_preparation/ # Corpus acquisition, subset extraction, and indexing
|   |-- alignment_preprocessing/ # MMS_FA alignment verification & text cleaners
|   |-- training_inference/  # Acoustic diagnostics and controlled mel tests
|   |-- plotting/            # plot_variance_adaptor.py (30 visualizations)
|   `-- verification/        # audit_submission.py & phase verification tests
|-- reference_fastspeech2/   # FastSpeech 2 core model, HiFi-GAN vocoder, and text modules
|-- requirements.txt         # Complete pinned dependencies
|-- .gitignore               # Configured ignore patterns for reproducible deployment
`-- README.md                # Comprehensive documentation and reproducibility guide
```